Summary in Graph

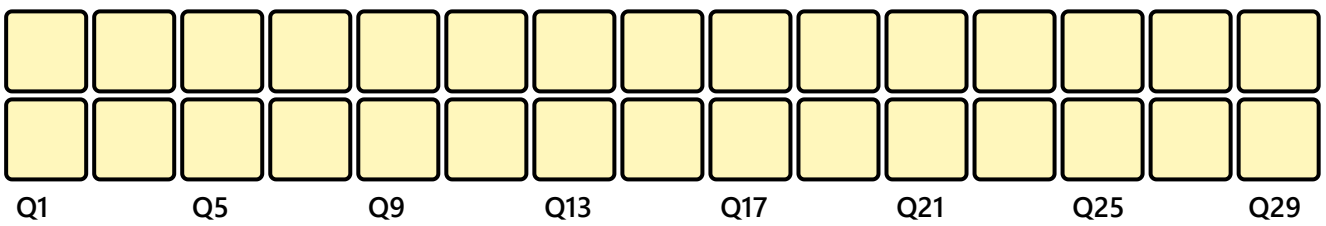
Exam Summary (GATE Overflow CS Test Series 2026 | Linear Algebra | Test 1)

Qs. Attempted:	0 (0 + 0)	Correct Marks:	0 (0 + 0)
Correct Attempts:	0 (0 + 0)	Penalty Marks:	0 (0 + 0)
Incorrect Attempts:	0 (0 + 0)	Resultant Marks:	0 (0 + 0)

Total Questions:	30 (10 + 20)
Total Marks:	50 (10 + 40)
Exam Duration:	90 Minutes
Time Taken:	0 Minutes

Individual Question Statistics

Select Metric: Marks Obtained



* Average Stats based on first-attempt stats of Top 10% (4) out of 43 users

EXAM RESPONSE | EXAM STATS | FEEDBACK

Technical

Q #1 | Numerical Type | Award: 1 | Penalty: 0 | Linear Algebra

Let $P = [p_{ij}]$, where $1 \leq i, j \leq n$ is defined by $p_{ij} = i \times j, \forall i, j$ with $n > 2$. The rank of P is _____

Your Answer: | Correct Answer: 1 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #2 Numerical Type | Award: 1 | Penalty: 0 | Linear Algebra

The rank of the matrix $A = \begin{bmatrix} 1 & 0 & 2 & 1 \\ 0 & 2 & 4 & 2 \\ 0 & 2 & 2 & 1 \end{bmatrix}$ is _____

Your Answer: | Correct Answer: 3 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #3 Numerical Type | Award: 1 | Penalty: 0 | Linear Algebra

The value of the determinant $\begin{vmatrix} 2\alpha & \alpha+\beta & \alpha+\gamma \\ \beta+\alpha & 2\beta & \beta+\gamma \\ \gamma+\alpha & \gamma+\beta & 2\gamma \end{vmatrix}$ is _____

Your Answer: | Correct Answer: 0 | Not Attempted | Time taken: 00min 02sec | Discuss

Q #4 Multiple Choice Type | Award: 1 | Penalty: 0.33 | Linear Algebra

Given a matrix $A = \frac{1}{2}\begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -1 & 1 & -1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & -1 & 1 \end{bmatrix}$, then $(AA^{-1})^T$ is _____

A. I
B. $\frac{1}{2}I$
C. $4I$
D. $\frac{1}{4}I$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #5 Multiple Select Type | Award: 1 | Penalty: 0 | Linear Algebra

If $A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$, the value of $(A^{-1})^T$ is ____
(Mark all the appropriate choices)

A. A
B. $Adj\ A$
C. $\left((A^T)^T\right)^T$
D. A^T

Your Answer: | Correct Answer: A;C;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #6 Multiple Select Type | Award: 1 | Penalty: 0 | Linear Algebra

Which of the following is/are correct? (Mark all the appropriate choices)

A. The value of a determinant remains unaltered, if the rows and columns are interchanged.

B. If two rows (or two columns) of a determinant are interchanged, the sign of the value of the determinant changes.
C. If two rows (or columns) of a determinant are identical, the value of the determinant is zero.
D. If the elements of any row (or column) of a matrix are multiplied by a constant k, the determinant also gets multiplied by k.

Your Answer: | Correct Answer: A;B;C;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #7 | Multiple Select Type | Award: 1 | Penalty: 0 | Linear Algebra

If A is an $n \times n$ non-singular matrix, such that $AA^T = A^TA$ and $B = A^{-1}A^T$, then BB^T is equal to ____ (Mark all the appropriate choices)

A. I
B. B^{-1}
C. $(B^{-1})^T$
D. I^T

Your Answer: | Correct Answer: A;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #8 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Linear Algebra

If A is skew- symmetric matrix of order 3, then the matrix A^6 is?

A. Diagonal matrix
B. Skew-symmetric matrix
C. Symmetric matrix
D. None of the above

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #9 | Multiple Select Type | Award: 1 | Penalty: 0 | Linear Algebra

Which of the following is/are correct? (Mark all the appropriate choices)

A. Addition of two symmetric matrices results in a symmetric matrix.
B. If A and B are two symmetric commutative matrices, i.e. $AB = BA$, then the product of A and B is symmetric.
C. If matrix A is symmetric then A^n is also symmetric, where n is any positive integer.
D. If A is an invertible symmetric matrix then A^{-1} is symmetric.

Your Answer: | Correct Answer: A;B;C;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #10 | Multiple Select Type | Award: 1 | Penalty: 0 | Linear Algebra

Which of the following is/are correct? (Mark all the appropriate choices)

A. The determinant of a skew-symmetric matrix over real numbers is non-negative.
B. When the identity matrix is added to a real skew-symmetric matrix, the resultant matrix is invertible.
C. For two matrices A and B, the equation $AB = 0$ implies that either A or B is a zero matrix.
D. For two matrices A and B, if $AB = 0$, then $BA = 0$.

Your Answer: | Correct Answer: A;B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #11 Numerical Type | Award: 2 | Penalty: 0 | Linear Algebra

If $A = \begin{bmatrix} 1 & -3 & -4 \\ -1 & 3 & 4 \\ 1 & -3 & -4 \end{bmatrix}$, is a nilpotent matrix, its index is _____

Your Answer: | Correct Answer: 2 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #12 Multiple Select Type | Award: 2 | Penalty: 0 | Linear Algebra

Let ω be the complex root of unity with $\omega \neq 1$, and $A = [a_{ij}]$ be a $n \times n$ matrix, with $a_{ij} = \omega^{i+j}$. Then $A^2 = O$, when n is equal to _____. (Mark all the appropriate choices)

A. 3691215
B. 1230403
C. 7865196
D. 4586325

Your Answer: | Correct Answer: A;C;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #13 Numerical Type | Award: 2 | Penalty: 0 | Linear Algebra

The determinant of the matrix $A = \begin{pmatrix} \frac{1}{2} & \alpha & 1 & 1 & 1 & 1 & 1 & 1 \\ 0 & -2 & 1 & 1 & 2 & -3 & 8 & -9 \\ 0 & 0 & -88 & 0 & 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & \frac{1}{22} & 5 & 4 & 3 & 21 \\ 0 & 0 & 0 & 0 & 0 & 1 & 5 & 0 \\ 0 & 0 & 0 & 0 & 0 & 110 & 0 & 47 \\ 0 & 0 & 0 & 0 & 0 & 0 & 48 & 78 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{4} \end{pmatrix}$ is _____

Your Answer: | Correct Answer: 0 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #14 Numerical Type | Award: 2 | Penalty: 0 | Linear Algebra

The value of $\begin{vmatrix} a^2 & a & bc \\ b^2 & b & ca \\ c^2 & c & ab \end{vmatrix} + \begin{vmatrix} 1 & 1 & 1 \\ a^2 & b^2 & c^2 \\ a^3 & b^3 & c^3 \end{vmatrix} =$ _____

Your Answer: | Correct Answer: 0 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #15 Multiple Select Type | Award: 2 | Penalty: 0 | Linear Algebra

If ω is the one of the imaginary cube roots of unity, the value of the determinant $\begin{vmatrix} 1 & \omega & \omega^2 \\ \omega & \omega^2 & 1 \\ \omega^2 & 1 & \omega \end{vmatrix} =$ ____

(Mark all the appropriate choices)

A. 1
B. ω^3
C. 0
D. $1+\omega+\omega^2$

Your Answer: | Correct Answer: C;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #16 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Linear Algebra

The determinant $\begin{vmatrix} \alpha p+\beta & \alpha & \beta \\ \beta p+\gamma & \beta & \gamma \\ 0 & \alpha p+\beta & \beta p+\gamma \end{vmatrix} = 0$ if

A. α, β, γ are in G.P.
B. α, β, γ are in A.P.
C. α, β, γ are in H.P.
D. $\alpha\beta, \beta\gamma, \gamma\alpha$ are in G.P.

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #17 | Numerical Type | Award: 2 | Penalty: 0 | Linear Algebra

If the matrix $P = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}$, then the value of $\det(\mathrm{P})$ is _____

Your Answer: | Correct Answer: -1 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #18 | Multiple Select Type | Award: 2 | Penalty: 0 | Linear Algebra

Consider the following system of equations

- $x_1 + x_2 - 2x_3 = 0$
- $x_1 - 2x_2 + x_3 = 0$
- $-2x_1 + x_2 + x_3 = 0$

The system has _____ (Mark all the appropriate choices)

A. infinite number of solutions
B. no solution
C. only a trivial solution
D. a unique solution

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #19 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Linear Algebra

The system of equations given below has ____

- $x + y + z + 3 = 0$
- $3x + y - 2z + 2 = 0$
- $2x + 4y + 7z - 7 = 0$

A. a unique solution
B. no solution
C. infinite number of solutions
D. exactly two solutions

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #20 | Multiple Select Type | Award: 2 | Penalty: 0 | Linear Algebra

Which of the following is/are correct? (Mark all the appropriate choices)

A. An $n \times n$ matrix is invertible if and only if it does not have 0 as an eigenvalue.
B. If a matrix A has eigenvalue λ with corresponding eigenvector x, then for any positive integer k, A^k has eigenvalue λ^k corresponding to the same eigenvector x.
C. If A is an invertible matrix with eigenvalue λ corresponding to eigenvector x, then A^{-1} has eigenvalue λ^{-1} corresponding to the same eigenvector x.
D. Eigenvectors of a matrix A with distinct eigenvalues are linearly independent.

Your Answer: | Correct Answer: A;B;C;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #21 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Linear Algebra

The value of $'x'$ for which all the eigen values of the matrix given below $\begin{bmatrix} 2 & 3+i & 6 \\ x & 3 & 2 \\ 6 & 2 & 3 \end{bmatrix}$ are real is _____

$(i = \sqrt{-1})$

A. $3 + i$
B. $1 - 3i$
C. $3 - i$
D. $1 + 3i$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #22 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Linear Algebra

The eigen values of $A = \begin{bmatrix} 3 & 3 & 3 & 3 & 3 \\ 3 & 3 & 3 & 3 & 3 \\ 3 & 3 & 3 & 3 & 3 \\ 3 & 3 & 3 & 3 & 3 \\ 3 & 3 & 3 & 3 & 3 \end{bmatrix}$ are _____

A. $0, 0, 0, 15, 0$
B. $0, 0, 0, 1, 14$
C. $1, 2, 3, 4, 5$
D. $3, 3, 3, 3, 3$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #23 Numerical Type | Award: 2 | Penalty: 0 | Linear Algebra

If the eigen value of a 2×2 matrix A are -3 and -4, the sum of the eigen value of the matrices $(A+2I)^{-1}$ and $(A+8I)^2 =$ _____

Your Answer: | Correct Answer: 39.5 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #24 Numerical Type | Award: 2 | Penalty: 0 | Linear Algebra

Let A be a real 2×2 matrix. If $7+5i$ is an eigenvalue of A, $\det(\mathrm{A}) = \alpha$, and $\mathrm{trace}(\mathrm{A}) = \beta$, then the value of $2\alpha + 3\beta$ is _____

Your Answer: | Correct Answer: 190 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #25 Multiple Select Type | Award: 2 | Penalty: 0 | Linear Algebra

Given the matrix $A = \begin{pmatrix} 6 & 5 \\ 5 & 6 \end{pmatrix}$, which of the following are the eigen vectors of A? (Mark all the appropriate choices)

A. $\begin{pmatrix} 1 \\ 1 \end{pmatrix}$ and $\begin{pmatrix} -5 \\ 5 \end{pmatrix}$

B. $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$ and $\begin{pmatrix} -5 \\ 0 \end{pmatrix}$

C. $\begin{pmatrix} -1 \\ 1 \end{pmatrix}$ and $\begin{pmatrix} 3 \\ 3 \end{pmatrix}$

D. $\begin{pmatrix} -1 \\ 0 \end{pmatrix}$ and $\begin{pmatrix} 0 \\ -5 \end{pmatrix}$

Your Answer: | Correct Answer: A;C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #26 Numerical Type | Award: 2 | Penalty: 0 | Linear Algebra

The value of $k^2 + 2k + 1$ such that the vector $\begin{bmatrix} 2 \\ 4 \\ 6 \end{bmatrix}$ is an eigenvector of the matrix $\begin{bmatrix} 4 & 1 & 2 \\ k & 2 & 1 \\ 14 & -4 & 10 \end{bmatrix}$ is _____

Your Answer: | Correct Answer: 324 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #27 Multiple Choice Type | Award: 2 | Penalty: 0.67 | Linear Algebra

The matrix $A = \begin{bmatrix} 3 & 2 & 2 \\ 2 & 3 & 2 \\ 2 & 2 & 3 \end{bmatrix}$ has three distinct eigenvalues and one of its eigenvectors is $\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$. Which one of the following can be another eigenvector of A?

A. $\begin{bmatrix} -1 \\ 1 \\ 1 \end{bmatrix}$

B. $\begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}$

C. $\begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}$

D. $\begin{bmatrix} -1 \\ 0 \\ -1 \end{bmatrix}$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #28

Multiple Select Type | Award: 2 | Penalty: 0 | Linear Algebra

Which of the following matrices do not have a LU decomposition? (Mark all the appropriate choices)

A. $\begin{bmatrix} 1 & 2 & 4 \\ 3 & 8 & 14 \\ 2 & 6 & 13 \end{bmatrix}$

B. $\begin{bmatrix} 1 & 2 & 4 \\ 4 & 8 & 14 \\ 2 & 6 & 13 \end{bmatrix}$

C. $\begin{bmatrix} 1 & 2 & 3 \\ 3 & 8 & 14 \\ 2 & 6 & 13 \end{bmatrix}$

D. $\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #29

Multiple Choice Type | Award: 2 | Penalty: 0.67 | Linear Algebra

P : The matrix $\begin{bmatrix} 0 & 4 \\ 0 & 5 \end{bmatrix}$ has infinitely many LU factorizations, where L is lower triangular and U is upper triangular with each diagonal entry 1.

Q : The matrix $\begin{bmatrix} 5 & 4 \\ 0 & 0 \end{bmatrix}$ has no LU factorization, where L is lower triangular and U is upper triangular with each diagonal entry 1.

For the given statements P and Q,

A. P is true and Q is false
B. Both P and Q are true
C. P is false and Q is true
D. Both P and Q are false

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #30 Numerical Type Award: 2 Penalty: 0 Linear Algebra

In the LU decomposition of the matrix $\begin{bmatrix} 3 & 5 \\ 6 & 8 \end{bmatrix}$, if the diagonal elements of U are both 1, then the lower diagonal entry l_{22} of L is _____

Your Answer: Correct Answer: -2 Not Attempted Time taken: 00min 00sec Discuss

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